Decent Coaching Classes

X SB

Mathematics: Part 1 (Ch. 1 - Linear Equations, Ch. 2 - Quadratic Equations) & Part 2 (Ch. 1 - Similarity, Ch. 2 - Pythagoras Theorem)

Time: 1 Hour Marks: 30

Q.1 (A) Choose the correct alternative. (5)

1) The standard form of a quadratic equation is:

a) $ax + b = 0$ b) $ax^2 + bx + c = 0$ ($a \neq 0$) c) $ax^2 + b = 0$ d) $ax + by = c$

2) For simultaneous equations in two variables, if $a1/a2 = b1/b2 \neq c1/c2$, the equations have:

a) a unique solution b) no solution c) infinitely many solutions d) only $x = 0$

3) In ΔABC, seg DE || side BC. If $AD/DB = 2/3$, then AE/EC = ?

a) 2/3 b) 3/2 c) 2/5 d) 3/5

4) In a right-angled triangle, the square of the hypotenuse is equal to:

a) the sum of squares of the other two sides b) the difference of squares of the other two sides

c) the product of the other two sides d) twice the sum of the other two sides

5) The discriminant of the quadratic equation $ax^2 + bx + c = 0$ is given by:

a) $b^2 - 4ac$ b) $b^2 + 4ac$ c) $4ac - b^2$ d) $2b - 4ac$

Q.2 Attempt any FOUR of the following. (4 × 2 = 8)

1) Solve the simultaneous equations: $x + y = 7$; $x - y = 3$

2) Solve using the formula method: $x^2 - 5x + 6 = 0$

3) In ΔPQR, seg XY || side QR. If PX = 2, XQ = 4, PY = 3, find YR.

4) Find the length of the diagonal of a square whose side is 6 cm.

5) Determine the nature of the roots of the quadratic equation: $2x^2 + 3x + 4 = 0$

Q.3 Attempt any THREE of the following. (3 × 3 = 9)

1) Solve the simultaneous equations: $2x + 3y = 11$; $2x - 4y = -24$

2) Form the quadratic equation whose roots are 4 and -3.

3) In ΔABC, $\angle B = 90°$. If AB = 5 cm and BC = 12 cm, find AC and the area of ΔABC.

4) State and prove the Basic Proportionality Theorem (BPT).

Q.4 Attempt any TWO of the following. (2 × 4 = 8)

1) The sum of the digits of a two-digit number is 9. If 27 is added to the number, the digits get reversed. Find the number.

2) The product of Rohan's age (in years) 5 years ago and his age 9 years later is 15. Find his present age.

3) In ΔPQR, $\angle Q = 90°$, seg QM is the altitude drawn to hypotenuse PR. If PM = 9 and MR = 16, find QM, PQ, and QR.

— All the Best —